

John Benedickson

Doylestown, PA | (610) 757-8773 | jbenedickson826@gmail.com

WORK EXPERIENCE

FreeTech, Remote

December 2023 – Present

Full-Stack Developer

- Architected a TypeScript monorepo (pnpm) with a shared domain package consumed by React, Cloud Functions, and Express — eliminating model drift across 3 client apps and backend services.
- Built a role-based React portal (Vite, MUI, TanStack Query) serving admins, clients, and transporters across inventory and sales workflows, loan application review, and logistics management.
- Developed Firebase/GCP backend: Firestore-triggered Cloud Functions for domain automation, custom in-app and email notifications, and an Express API on Cloud Run with Stripe, Shippo, and PandaDoc integrations.
- Owned Firestore security rules, composite indexes, and TypeScript migration scripts across staging and production, enabling schema and role changes without service disruption.

Newport News Shipbuilding, Newport News, VA

July 2023 – July 2025

Software Engineer, Instrumentation & Control

- Partnered with a third-party vendor to deliver an LLM-based RAG tool enabling engineers to query ship specification documents in natural language, reducing time spent searching technical documentation across the program.
- Developed an interactive HMI prototype for the electric plant control panel in Visio/VBA, supporting design review and operator validation of control logic.
- Programmed a Xilinx Zynq UltraScale+ ZCU102 to emulate commercial PLC behavior for shipboard I/O control as the sole engineer on the effort, learning the Vivado/Vitis toolchain independently and authoring the reference documentation used to onboard the next engineer.

SKILLS

Languages: TypeScript, JavaScript, C

Frameworks: React, Express, MUI, TanStack Query, Node.js

Platforms: Firebase, Google Cloud Platform, Cloud Run

Concepts: REST APIs, CI/CD

EDUCATION

West Virginia University, Morgantown, WV

May 2023

Bachelor of Science in Computer Engineering (ABET-accredited)